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## Developing cartridge


### Abstract

A developing cartridge includes a developing frame, a developing roller, a first drive force receiving member, and a first end cover. The first drive force receiving member is configured to receive a driving force to drive the rotation of the developing roller. The developing cartridge further includes a conductive assembly and a first voltage control member. The conductive assembly is capable of switching internally between an electrically connected state and an electrically non-connected state, and the conductive assembly is provided with a first electrical connection portion that is elastically deformable. The first voltage control member is rotatably mounted on the first end cover and configured to move in response to movement of the first drive force receiving member, thereby switching the conductive assembly between the electrically connected state and the electrically non-connected state, thereby facilitating the miniaturization of the developing cartridge.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of International Application No. PCT/CN2024/123329, filed on Oct. 8, 2024, which claims priority to Chinese Patent Application No. 202322692719.3, filed on Oct. 8, 2023; Chinese Patent Application No. 202323366031.2, filed on Dec. 8, 2023; and Chinese Patent Application No. 202421499471.7, filed on Jun. 27, 2024. All of the aforementioned applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The present disclosure relates to the field of electrophotographic imaging devices, and particularly to a developing cartridge.
BACKGROUND
(2) As is known in the art, imaging devices include a main assembly for mounting developing cartridges/drum cartridges and a detection device for determining information of the developing cartridge installed in the imaging device. For example, the detection device is used to determine whether a developing cartridge newly installed in the imaging device is a new product or to determine information such as the size and capacity of the developing cartridge. This type of existing imaging device can be referred to the contents disclosed in U.S. Pat. No. 10,025,225B2 and its family patents.
(3) An existing developing cartridge detachably installable in such imaging devices incorporates a first drive force receiving member and a detectable device. The first drive force receiving member rotates upon receiving driving force from the imaging device, while the detectable device moves in response to this rotation of the first drive force receiving member. The detectable device can actuate the switching member of the detection device to move between closed and open positions, thus changes the electrical signals of internal circuit of the detection device for detection of the developing cartridge. However, the developing cartridge with this structure requires numerous transmission mechanisms, resulting in complexity and impeding miniaturization of the developing

cartridge. Furthermore, the detection process of the developing cartridge with this structure generates mechanical collision noise, degrading user experience.

(4) In the research of such developing cartridges, key ongoing technical challenges for those skilled in the art include simplifying developing cartridge structure and implementing developing cartridge detection methods.

SUMMARY

(5) The objective of the present disclosure is to provide a developing cartridge that advances the aforementioned technology.

(6) The first aspect of the present disclosure discloses a developing cartridge, comprising: a developing frame internally provided with a developer chamber configured to accommodate developer; a developing roller rotatably disposed on the developing frame; a first drive force receiving member configured to receive a driving force for driving rotation of the developing roller; a first end cover, connected to the developing frame, to at least partially cover the first drive force receiving member; a conductive assembly capable of switching internally between an electrically connected state and an electrically non-connected state, wherein the conductive assembly is provided with a first electrical connection portion that is elastically deformable; and a first voltage control member mounted on the first end cover and configured to move in response to movement of the first drive force receiving member, thereby switching the conductive assembly between the electrically connected state and the electrically non-connected state.

(7) Furthermore, the developing cartridge is detachably installable in an imaging device, the imaging device further comprises an openable door, the door having a groundable first inner wall, and the first electrical connection portion is configured to electrically contact the first inner wall.

(8) Furthermore, the first electrical connection portion comprises an anti-slip portion, the anti-slip portion abuts the first inner wall, and at least a portion of the anti-slip portion is made of an elastic conductive material.

(9) Furthermore, the first electrical connection portion is configured to move in a front-rear direction of the developing cartridge.

(10) Furthermore, the first electrical connection portion is configured to move between a first position and a second position under an external force, wherein in the first position, the first electrical connection portion protrudes relative to the developing frame, and in the second position, the first electrical connection portion is retracted relative to the first position.

(11) Furthermore, the first electrical connection portion is disposed on the first end cover, and viewed from a front-rear direction of the developing cartridge, the first electrical connection portion at least partially overlaps with the first voltage control member.

(12) Furthermore, along a longitudinal direction of the developing cartridge, the developing cartridge has a first end portion and an opposing second end portion, the conductive assembly comprises a first conductive member, a second conductive member, a third conductive member, and a fourth conductive member, the first conductive member is disposed at the second end portion, the second conductive member is electrically connected to the first conductive member, the third conductive member is electrically connected to the second conductive member and configured to connect to the fourth conductive member; at least a portion of the second conductive member extends between the first end portion and the second end portion of the developing cartridge, at least a portion of the third conductive member is disposed at the first end portion of the developing cartridge; the fourth conductive member includes the first electrical connection portion; the first voltage control member comprises a first main body portion, a first drive force receiving portion, and a first driving portion, the first drive force receiving portion is disposed on the first main body portion and configured to receive a driving force; the first driving portion is disposed on the first main body portion and is capable of driving the third conductive member or the fourth conductive member so that the third conductive member and the fourth conductive member is capable of switching between the electrically connected state and the electrically non-connected state.

(13) Furthermore, the first conductive member, the second conductive member, the third conductive member and the fourth conductive member are separately formed or at least two are integrally formed.
(14) Furthermore, the fourth conductive member includes a first sub-member and the first electrical connection portion, and the first sub-member is provided with a first sub-gear.
(15) Furthermore, the developing cartridge further comprises a supply roller rotatably supported on the developing frame and configured to supply developer to the developing roller, wherein the second conductive member is the supply roller.
(16) The second aspect of the present disclosure discloses a developing cartridge, comprising: a developing frame internally provided with a developer chamber configured to accommodate developer; a developing roller rotatably disposed on the developing frame; a first drive force receiving member configured to receive a driving force for driving rotation of the developing roller; a first end cover, connected to the developing frame, to at least partially cover the first drive force receiving member; a conductive assembly capable of switching internally between an electrically connected state and an electrically non-connected state, wherein the conductive assembly is provided with a first electrical connection portion that is elastically deformable; and a first voltage control member configured to move in response to movement of the first drive force receiving member, thereby switching the conductive assembly between the electrically connected state and the electrically non-connected state.
(17) Furthermore, the developing cartridge is detachably installable in a drum cartridge having a photosensitive drum, the drum cartridge is detachably installable in an imaging device, the drum cartridge comprises a separation force transmission member, and the developing cartridge comprises a separation force receiving member configured to receive an acting force from the separation force transmission member or a separation force applying member of the imaging device to separate the developing roller from the photosensitive drum, wherein before separation of the developing roller from the photosensitive drum, the first electrical connection portion has a first deformation amount, and after separation of the developing roller from the photosensitive drum, the first electrical connection portion has a second deformation amount, wherein the first deformation amount is greater than the second deformation amount.
(18) Furthermore, the developing cartridge is detachably installable in a drum cartridge having a photosensitive drum, the drum cartridge is detachably installable in a main assembly of an imaging device, and the main assembly comprises a main frame and a mounting cavity configured to mount the developing cartridge, wherein the main frame comprises a first upper wall and a second upper wall, wherein the second upper wall is closer to an inside of the mounting cavity than the first upper wall, wherein the first upper wall is made of a conductive material, the second upper wall is made of a non-conductive material, and the first electrical connection portion is configured to electrically connect with the first upper wall.
(19) Furthermore, the first upper wall comprises a plurality of labels, wherein there is a first gap between the labels and the second upper wall, there is a second gap between the plurality of labels, and the first electrical connection portion makes electrical contact with the first gap or the second gap.
(20) Furthermore, the developing cartridge is detachably installable in a drum cartridge having a photosensitive drum, the drum cartridge is detachably installable in an imaging device, and the developing cartridge further comprises a storage unit having a ground terminal capable of electrically connecting with a contact pin of the imaging device, and the first electrical connection portion is capable of electrically connecting with the ground terminal.
(21) Furthermore, along a longitudinal direction of the developing cartridge, the developing cartridge has a first end portion and an opposing second end portion, the conductive assembly comprises a first conductive member, a second conductive member, a third conductive member, and a fourth conductive member, the first conductive member is disposed at the second end portion,

the second conductive member is electrically connected to the first conductive member, the third conductive member is electrically connected to the second conductive member and configured to connect to the fourth conductive member; at least a portion of the second conductive member extends between the first end portion and the second end portion of the developing cartridge, at least a portion of the third conductive member is disposed at the first end portion of the developing cartridge; the fourth conductive member includes the first electrical connection portion; the first voltage control member comprises a first main body portion, a first drive force receiving portion, and a first driving portion, the first drive force receiving portion is disposed on the first main body portion and configured to receive a driving force; the first driving portion is disposed on the first main body portion and is capable of driving the third conductive member or the fourth conductive member so that the third conductive member and the fourth conductive member is capable of switching between the electrically connected state and the electrically non-connected state.
(22) Furthermore, the first conductive member, the second conductive member, the third conductive member and the fourth conductive member are separately formed or at least two are integrally formed.
(23) Furthermore, the imaging device comprises a detection device and a grounding terminal, the detection device comprises a detection unit, a power source and a switching member capable of forming a conductive circuit, wherein the switching member is movable between a closed position and an open position, and the conductive assembly is capable of electrically connecting the switching member and the grounding terminal.
(24) Furthermore, the developing cartridge further comprises a first force applying portion disposed on the first conductive member or formed separately from the first conductive member, wherein the first force applying portion is configured to apply a force to a switching member of the detection device when the developing cartridge is installed in the imaging device to move the switching member from an initial position to a closed position.
(25) Furthermore, the developing cartridge further comprises a deceleration mechanism disposed at a first end portion of the developing cartridge, wherein the deceleration mechanism comprises at least a first transmission member and a second transmission member, wherein the first transmission member comprises a third gear portion and a fourth gear portion, a diameter of the third gear portion is smaller than that of the fourth gear portion, wherein the second transmission member comprises a fifth gear portion and a second gear portion, the fourth gear portion meshes with a first gear portion of the first drive force receiving member, the third gear portion meshes with the fifth gear portion, and the second gear portion meshes with a first drive force receiving portion of the first voltage control member.
(26) The developing cartridge with the above structure features simplified construction that facilitates the miniaturization of the developing cartridge.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) One or more embodiments are illustrated by means of the corresponding drawings, which do not constitute a limitation of the embodiments, connection ports having the same reference numerals in the drawings are shown as similar connection ports, and the drawings do not constitute a limitation of proportion. In the drawings:
(2) FIG. **1** is a schematic diagram showing the internal circuit of the detection device when the switching member is in the open position (initial position) in the present disclosure;
(3) FIG. **2** is a schematic diagram showing the internal circuit of the detection device when the switching member is in the closed position in the present disclosure;
(4) FIG. **3** is a schematic diagram showing the internal circuit of the detection device when the

switching member is in the open position in the present disclosure;
(5) FIG. **4** is a structural view showing the developing cartridge of Embodiment 1 installed in the imaging device in the present disclosure;
(6) FIG. **5***a* is a simplified structural view of the imaging device in Embodiment 1 in the present disclosure;
(7) FIG. **5***b* is a structural view showing the switching member and swing arm in Embodiment 1 in the present disclosure;
(8) FIG. **6** is a structural view showing the developing cartridge of Embodiment 1 installed in the drum cartridge in the present disclosure;
(9) FIG. **7** is a structural view showing another angle of the developing cartridge of Embodiment 1 installed in the drum cartridge in the present disclosure;
(10) FIG. **8** is a structural view of the developing cartridge of Embodiment 1 in the present disclosure;
(11) FIG. **9** is a partial structural view of one end of the developing cartridge of Embodiment 1 in the present disclosure;
(12) FIG. **10** is a partial structural view of the other end of the developing cartridge of Embodiment 1 in the present disclosure;
(13) FIG. **11** is a structural view showing another angle of the developing cartridge of Embodiment 1 in the present disclosure;
(14) FIG. **12** is a schematic diagram showing the internal circuit of the detection device when the developing cartridge of Embodiment 1 is not installed in the imaging device in the present disclosure;
(15) FIG. **13** is a schematic diagram showing the circuit formed between the detection device and detectable device when the developing cartridge of Embodiment 1 is installed in the imaging device in the present disclosure;
(16) FIG. **14** is a schematic diagram showing the circuit when the detection device and grounding terminal are conducting while the developing cartridge of Embodiment 1 is installed in the imaging device in the present disclosure;
(17) FIG. **15** is a structural view showing the developing cartridge of Embodiment 2 installed in the imaging device in the present disclosure;
(18) FIG. **16** is a structural view showing the developing cartridge of Embodiment 3 before installation in the imaging device in the present disclosure;
(19) FIG. **17** is a structural view showing the developing cartridge of Embodiment 3 installed in the imaging device in the present disclosure;
(20) FIG. **18** is a structural view of the developing cartridge of Embodiment 4 in the present disclosure;
(21) FIG. **19** is a schematic view of the developing cartridge of Embodiment 4 with certain structures hidden in the present disclosure;
(22) FIG. **20** is a schematic diagram showing the internal circuit of the detection device when the developing cartridge of Embodiment 4 is not installed in the imaging device in the present disclosure;
(23) FIG. **21** is a schematic diagram showing the circuit formed between the detection device and detectable device when the developing cartridge of Embodiment 4 is installed in the imaging device in the present disclosure;
(24) FIG. **22** is a schematic diagram showing the circuit when the detection device and grounding terminal are conducting while the developing cartridge of Embodiment 4 is installed in the imaging device in the present disclosure;
(25) FIG. **23** is a structural view of the developing cartridge of Embodiment 5 in the present disclosure;
(26) FIG. **24** is a partial structural view of one end of the developing cartridge of Embodiment 5 in

the present disclosure;
(27) FIG. **25** is a partial structural view of the other end of the developing cartridge of Embodiment 5 in the present disclosure;
(28) FIG. **26** is a structural view of the developing cartridge of Embodiment 5 with certain structures hidden in the present disclosure;
(29) FIG. **27** is a schematic diagram showing the internal circuit of the detection device when the developing cartridge of Embodiment 5 is not installed in the imaging device in the present disclosure;
(30) FIG. **28** is a schematic diagram showing the circuit formed between the detection device and detectable device when the developing cartridge of Embodiment 5 is installed in the imaging device in the present disclosure;
(31) FIG. **29** is a schematic diagram showing the circuit when the detection device and grounding terminal are conducting while the developing cartridge of Embodiment 5 is installed in the imaging device in the present disclosure;
(32) FIG. **30** is a structural view of the developing cartridge of Embodiment 6 in the present disclosure;
(33) FIG. **31** is a structural view showing another angle of the developing cartridge of Embodiment 6 in the present disclosure;
(34) FIG. **32** is an exploded partial structural view of the developing cartridge of Embodiment 6 in the present disclosure;
(35) FIG. **33** is an overall structural view of the developing cartridge of Embodiment 7 in the present disclosure;
(36) FIG. **34** is an exploded partial structural view of the developing cartridge of Embodiment 7 in the present disclosure;
(37) FIG. **35** is a structural view showing the assembled state of the first end cover, movable member, third conductive member and fourth conductive member of the developing cartridge of Embodiment 7 in the present disclosure;
(38) FIG. **36** is a structural view showing the disassembled state of the first end cover, movable member, third conductive member and fourth conductive member of the developing cartridge of Embodiment 7 in the present disclosure;
(39) FIG. **37** is a partial structural view of one end of the developing cartridge of Embodiment 8 in the present disclosure;
(40) FIG. **38** is a partial structural view of the third conductive member and fourth conductive member of Embodiment 8 in the present disclosure; and
(41) FIG. **39** is a partial structural view showing the third conductive member and fourth conductive member of a variation of Embodiment 8 in the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(42) The present disclosure will be described in further detail with reference to the embodiments and accompanying drawings.

Embodiment 1

(43) As shown in FIGS. **1-14**, the present embodiment provides an imaging device P comprising a main assembly P**1**, a door P**2**, a detection device **50**, and a separation force applying member (not shown in the figures). The main assembly P**1** comprises a mounting cavity P**3** and a main frame P**4**. The mounting cavity P**3** is configured to mount the developing cartridge **10**, and comprises a second inner wall P**31** with a plurality of grounding conductive members P**311** exposed toward the inside of the mounting cavity P**3** from the inner wall.
(44) The main frame P**4** surrounds the mounting cavity P**3** and has at least a portion made of conductive material. The main frame P**4** comprises a conductive and groundable first upper wall P**41** and a non-conductive second upper wall P**42**. The second upper wall P**42** is positioned closer to the inside of the mounting cavity P**3** than the first upper wall P**41**. The first upper wall P**41** is

made of conductive metal material, while the second upper wall P**42** is made of non-conductive resin. The second upper wall P**42** comprises a first surface P**43** and a second surface P**44** forming a height difference, with the second surface P**44** being further from the first upper wall P**41** compared to the first surface P**43**. The door P**2** can open and close the mounting cavity P**3** and is rotatably mounted on the main frame P**4**. The door P**2** comprises a conductive and groundable first inner wall P**21**.
(45) In some embodiments, labels P**47** are attached to the first upper wall P**41** for recording information about installation and use of the developing cartridge. A second gap P**472** is formed between adjacent labels P**47**, and a first gap P**471** is formed between the labels P**47** and the second upper wall P**42**.
(46) The detection device **50** comprises a detection unit **501**, a power source **502**, a switching member **503**, and a processor (not shown). The detection unit **501**, power source **502**, and switching member **503** can be electrically connected to form a conductive circuit, with the switching member **503** controlling circuit making and breaking. The detection unit **501** can detect changes in electrical signals based on circuit making and breaking, transmit different electrical signals to the processor, and thereby determine basic information about the developing cartridge. The switching member **503**, at least partially constructed of conductive material, is configured as a movable member with three positions: initial position, closed position, and open position. As shown in FIG. **1**, in the initial position (also referred to as the open position), the switching member **503** maintains a non-conducting circuit state. In the closed position (FIG. **2**), the switching member **503** closes the circuit, enabling conduction and allowing the detection unit **501** to detect electrical signals. In the open position (FIG. **3**), the switching member **503** returns to a non-conducting circuit state. As the movable member transitions between closed and open positions, the electrical signal (such as voltage) of the detection unit **501** changes, enabling determination of the information of developing cartridge.
(47) In some embodiments, the switching member **503** may be omitted, with the detection unit electrically connected to a conductive part of the developing cartridge (such as the detectable device to be described below), thereby affecting the detection unit's electrical signal through voltage changes in the detectable device. This allows the processor to determine basic information about the developing cartridge based on different electrical signals.
(48) In other embodiments, the detection unit may be a capacitive sensor, pressure sensor, photoelectric sensor, electromagnetic sensor or the like.
(49) In the present embodiment, the switching member **503** is made of conductive material and is configured as a movable member having a closed position and an open position. Specifically, at least a portion of the switching member **503** is disposed on a swing arm **504**, allowing the switching member **503** to move between the closed and open positions as the swing arm **504** pivots. As shown in FIG. **1**, in the open position (which may also serve as the initial position), the switching member **503** is in a non-conducting state. In the closed position (as shown in FIG. **2**), the switching member **503** closes the circuit, causing the circuit to be conducted and allowing the detection unit **501** to detect electrical signals. The separation force applying member (not shown) is disposed in the main assembly P**1** and can apply an acting force to developing cartridge **10**.
(50) As shown in FIGS. **4-6**, the developing cartridge **10** is detachably installable in a drum cartridge **20** having a photosensitive drum **201**, with the drum cartridge **20** being detachably installable in the main assembly P**1** of the imaging device P. For ease of description, the direction generally parallel to the installation direction of the drum cartridge **20** into the electrophotographic imaging device P is defined as the front-rear direction, with the side having the photosensitive drum **201** being the front side and the opposite side being the rear side. The direction parallel to the longitudinal direction of the drum cartridge **20** is defined as the left-right direction, with the side receiving driving force from the electrophotographic imaging device P being the left side and its opposite side being the right side. The direction perpendicular to both the front-rear direction and

left-right direction is defined as the up-down direction.

(51) As shown in FIGS. **6-8**, the drum cartridge **20** comprises a drum frame **202**, photosensitive drum **201**, second drive force receiving member **203**, charging member **204**, first electrode **205**, separation force transmission member **206**, pressing member, and locking member **207**. The drum frame **202** comprises a mounting portion for mounting the developing cartridge **10**. The photosensitive drum **201** is rotatably mounted on the drum frame **202**. The second drive force receiving member **203** is positioned at one side of the drum cartridge **20** to receive driving force from the imaging device P and directly or indirectly drive rotation of the photosensitive drum **201**. The charging member **204** charges the photosensitive drum **201**, and the first electrode **205** can receive first electrical power from the imaging device P and transmit it to the charging member **204**. The separation force transmission member **206** is disposed on the drum frame **202** and can move upon receiving force from the separation force applying member to transmit separation force to the developing cartridge **10**. In this embodiment, the separation force transmission member **206** is configured as a rotating member. The pressing member can apply pressing force to the developing cartridge **10** to bring the developing roller **13** mounted on the developing cartridge **10** into contact with the photosensitive drum **201**. The locking member **207** locks the developing cartridge **10** in position.

(52) As shown in FIGS. **7-11**, the developing cartridge **10** comprises a developing frame **11**, first drive force receiving member **12**, developing roller **13**, and storage unit **14**. Some developing cartridges **10** may also include a supply roller **15** and/or an agitation member **16**.

(53) Along the longitudinal direction of the developing cartridge **10**, the developing cartridge **10** has a first end portion **101** and an opposing second end portion **102**. The developing frame **11** comprises a developer chamber for accommodating developer. The developing roller **13** comprises a developing layer **131** and a developing roller shaft **132**, with the developing layer **131** wrapped around the outside of the developing roller shaft **132**. Both the developing roller shaft **132** and developing layer **131** can be electrically conductive. Viewed along the installation direction of the developing cartridge **10** into the imaging device P, the developing roller **13** is rotatably mounted on the front side of the developing frame **11**, exposed from the front side of the developing frame **11**, and positioned to face the photosensitive drum **201** when installed in the imaging device P to perform development operations. The storage unit **14** stores information such as specifications and model of the developing cartridge **10**. It comprises a substrate **141**, memory, and electrical contacts. The substrate **141** has a front surface and an opposing back surface. The electrical contacts and the memory are both mounted on the substrate **141**, with the electrical contacts mounted on the front surface of the substrate **141** and the memory mounted on the back surface of the substrate **141**. The electrical contacts include a plurality of contacts (two or more), including at least one ground terminal **142** that can electrically connect with the imaging device P to achieve grounding. In this embodiment, the storage unit **14** is disposed at the first end portion **101** of the developing cartridge **10**.

(54) The supply roller **15** is rotatably supported on the developing frame **11** facing the developing roller **13** and can supply developer to the developing roller **13**. The agitation member **16** agitates developer contained in the developer chamber. The first drive force receiving member **12** receives driving force from the imaging device P to directly or indirectly rotate the developing roller **13** and supply roller **15**. Preferably, the first drive force receiving member **12** is disposed at the first end portion **101** of the developing cartridge **10**.

(55) The first drive force receiving member **12** comprises a first force receiving portion **121**, second main body portion **122**, first gear portion **123**, and sixth gear portion **124**. The first force receiving portion **121** is disposed on the second main body portion **122** and can mesh with and receive driving force from a drive force output member of the imaging device P. The first gear portion **123** and sixth gear portion **124** connect to the second main body portion **122** and can transmit the driving force. The first gear portion **123** and sixth gear portion **124** may be integrally

formed with or movably connected to the second main body portion **122**. Preferably, the diameter of the second main body portion **122** is smaller than that of the first gear portion **123**, which is in turn smaller than that of the sixth gear portion **124**. The first gear portion **123** is positioned further from the developing frame **11** compared to the sixth gear portion **124**. The storage unit **14** may be disposed at the first end portion **101**, second end portion **102**, or at a position between the first end portion **101** and second end portion **102**. In this embodiment, the developing cartridge **10** also comprises a first end cover **103** positioned at the first end portion **101** of the developing cartridge **10**, connected to the developing frame **11**, and at least partially covering the first drive force receiving member **12**.
(56) This embodiment of the developing cartridge **10** further comprises a pressed member **17**, separation force receiving member **18**, and lockable member **19**. The pressed member **17** can receive force from the pressing member to bring the developing roller **13** and photosensitive drum **201** into proximity or contact. The separation force receiving member **18** receives force from the separation force transmission member **206**/separation force applying member to separate the developing roller **13** from the photosensitive drum **201**.
(57) The pressed member **17** is disposed on the developing frame **11**, preferably integrally formed with the developing frame **11**. When the developing cartridge **10** is installed in the drum cartridge **20**, the pressing member abuts against the pressed member **17** and applies force to the pressed member **17** from rear to front. More preferably, the pressed member **17** is positioned between the two end portions of the developing cartridge **10**. The separation force receiving member **18** is disposed at one end of the developing cartridge **10**. When the imaging device P executes a cleaning operation, the separation force transmission member **206**/separation force applying member applies force to the separation force receiving member **18**, causing the developing cartridge **10** to move rearward, thereby separating the developing roller **13** from the photosensitive drum **201**. Preferably, the separation force receiving member **18** is disposed on and protrudes from the developing frame **11**.
(58) In some embodiments, the separation force receiving member **18** is disposed on the second main body portion **122** of the first drive force receiving member **12**. The separation force receiving member **18** may be integrally formed with or separately formed from the second main body portion **122**. The lockable member **19** is disposed on the rear side of the developing cartridge **10**. When the developing cartridge **10** is installed in the drum cartridge **20**, the lockable member **19** engages with the locking member **207** to lock the developing cartridge **10** in position. Specifically, the lockable member **19** is disposed on the second end cover **104**. Preferably, the locking member **19** is positioned adjacent to the pressed member **17**.
(59) As shown in FIGS. **7-11**, this embodiment of the developing cartridge **10** also comprises a detectable device **70** that can be detected by the detection device **50** of the imaging device P.
(60) The detectable device **70** comprises a conductive assembly **72** and a first voltage control member **71**. When installed in the imaging device P, the conductive assembly **72** can electrically connect with the detection device **50**/switching member **503** and the grounding terminal K of the imaging device P. The first voltage control member **71** controls the making and breaking of electrical connection between the detection device **50** and grounding terminal K, thereby causing changes in voltage detected by the detection unit **501**. Specifically, the conductive assembly **72** comprises a first electrical connection portion **73** for electrical contact with the grounding terminal K. The grounding terminal K may be at least one of the following elements: the ground terminal **142** of the storage unit **14**, the imaging device contact pin that contacts the ground terminal **142**, the inner wall of the door P**2**, the first upper wall P**41**, and the grounding conductive members P**311**.
(61) In this embodiment, the first electrical connection portion **73** preferably contacts the first upper wall P**41**. In some embodiments, the first electrical connection portion **73** makes contact within the first gap P**471** or second gap P**472**.
(62) The conductive assembly **72** comprises a first conductive member **74**, second conductive

member **75**, third conductive member **76** and fourth conductive member **77**. The first conductive member **74**, second conductive member **75**, third conductive member **76** and fourth conductive member **77** can be electrically connected in sequence and may be separately formed or integrally formed.

(63) In this embodiment, at least a portion of the first conductive member **74** is disposed at the second end portion **102** of the developing cartridge **10** and can receive electrical power from the imaging device P, specifically through contact with the switching member **503** of the imaging device P. The second conductive member **75** electrically connects with the first conductive member **74** and has at least a portion extending between the first end portion **101** and second end portion **102** of the developing cartridge **10**. The second conductive member **75** may be at least one of the developing roller shaft **132**, developing roller **13**, supply roller **15**, and doctor blade **25**. In this embodiment, the developing roller shaft **132** serves as the second conductive member **75**. The third conductive member **76** electrically connects with the second conductive member **75** and has at least a portion disposed at the first end portion **101** of the developing cartridge **10**. The fourth conductive member **77** can selectively electrically connect with the third conductive member **76** and can electrically connect with the grounding terminal K, preferably connecting with the first upper wall P**41**. At least a portion of the fourth conductive member **77** is disposed at the first end portion **101** of the developing cartridge **10**.

(64) The conductive assembly **72** can switch internally between an electrically connected state and an electrically non-connected state. Specifically, the third conductive member **76** and fourth conductive member **77** can switch between an electrically connected state and an electrically non-connected state. At least a portion (the moving end) of the third conductive member **76** and/or fourth conductive member **77** is movable to enable switching between contact and non-contact states between the third conductive member **76** and fourth conductive member **77**. Preferably, the fourth conductive member **77** comprises a first electrical connection portion **73** that contacts the grounding terminal K, with this first electrical connection portion **73** being elastically deformable. When the developing cartridge **10** is installed in the drum cartridge **20**, the first electrical connection portion **73** abuts against the grounding terminal K. In some embodiments, when the developing cartridge **10** moves rearward under force from the separation force transmission member **206**/separation force applying member, the first electrical connection portion **73** maintains contact with the grounding terminal K, effectively improving contact stability between the first electrical connection portion **73** and grounding terminal K. In some embodiments, when the developing cartridge **10** is not under force from the separation force transmission member **206**/separation force applying member, the first electrical connection portion **73** has a first deformation amount, and when the developing cartridge moves rearward under force from the separation force transmission member **206**/separation force applying member, the first electrical connection portion **73** has a second deformation amount, with the first deformation amount being greater than the second deformation amount.

(65) Preferably, the first voltage control member **71** is configured to move between a third position and a fourth position. In the third position, the first voltage control member **71** drives the third conductive member **76** and/or fourth conductive member **77** to place the third conductive member **76** and fourth conductive member **77** in an electrically connected state. In the fourth position, the first voltage control member **71** does not drive the third conductive member **76** and/or fourth conductive member **77**, leaving the third conductive member **76** and fourth conductive member **77** in an electrically non-connected state.

(66) The first voltage control member **71** is configured to move in response to movement of the first drive force receiving member **12**. It can directly or indirectly drive at least a portion of the third conductive member **76** and/or fourth conductive member **77** to enable switching between contact and non-contact states between the third conductive member **76** and fourth conductive member **77**.

(67) The first voltage control member **71** is rotatably connected directly or indirectly to the developing frame **11**. In this embodiment, the first voltage control member **71** is rotatably mounted on the first end cover **103** of the developing cartridge **10**, thereby not only optimizing the installation space of the first end portion **101** for the assembly of the developing cartridge **10** but also allowing the first voltage control member **71** to be installed on different developing cartridges **10** by removing and installing the first end cover **103**, thereby improving the adaptability of the first voltage control member **71**. Meanwhile, the first voltage control member **71** is configured as a rotary member including a first main body portion **710**, first drive force receiving portion **711**, and first driving portion **712**. The first drive force receiving portion **711** is disposed on the first main body portion **710** for engaging with and receiving driving force from the first drive force receiving member **12**. The first driving portion **712** is disposed on the first main body portion **710** and can drive the third conductive member **76** to make electrical contact between the third conductive member **76** and fourth conductive member **77**.
(68) In this embodiment, in the longitudinal direction of the developing cartridge (i.e., left-right direction), the first voltage control member **71** at least partially overlaps with the storage unit **14**.
(69) The developing cartridge **10** of this embodiment also comprises a first force applying portion **105** that can apply force to the switching member **503** of the imaging device P when the developing cartridge **10** is installed in the imaging device P, moving the switching member **503** from its initial position to a closed position. In this embodiment, the first force applying portion **105** is disposed on the first conductive member **74**. Alternatively, the first force applying portion **105** may be disposed separately from the first conductive member **74**.
(70) The developing cartridge **10** of this embodiment further comprises a deceleration mechanism disposed at the first end portion **101** of the developing cartridge **10**. The deceleration mechanism comprises at least a first transmission member **81** and a second transmission member **82**. The first transmission member **81** comprises a third gear portion **811** and a fourth gear portion **812**, with the diameter of the third gear portion **811** being smaller than that of the fourth gear portion **812**. The third gear portion **811** is positioned closer to the developing frame **11** compared to the fourth gear portion **812**.
(71) The second transmission member **82** comprises a fifth gear portion **821** and a second gear portion **822**. The diameter of the fifth gear portion **821** is greater than that of the second gear portion **822**, with the fifth gear portion **821** being positioned closer to the developing frame **11** compared to the second gear portion **822**.
(72) The fourth gear portion **812** can mesh with the first gear portion **123** of the first drive force receiving member **12**, the third gear portion **811** can mesh with the fifth gear portion **821** of the second transmission member **82**, and the second gear portion **822** of the second transmission member **82** can mesh with the first drive force receiving portion **711** of the first voltage control member **71**. Preferably, the diameter of the third gear portion **811** is smaller than the diameters of both the first gear portion **123** and sixth gear portion **124**. Preferably, the diameter of the second gear portion **822** is smaller than that of the third gear portion **811**. Preferably, the diameter of the first drive force receiving portion **711** of the first voltage control member **71** is greater than that of the second gear portion **822**.
(73) Since the diameter of the third gear portion **811** is smaller than that of the fifth gear portion **821**, when the third gear portion **811** transmits driving force to the fifth gear portion **821**, the rotational angular velocity of the second transmission member **82** is less than that of the first transmission member **81**. Since the diameter of the second gear portion **822** is smaller than that of the first drive force receiving portion **711**, when the second transmission member **82** transmits driving force to the first voltage control member **71**, the rotational angular velocity of the first voltage control member **71** is less than that of the second transmission member **82**. Thus, when driving force is transmitted from the first drive force receiving member **12** to the first voltage control member **71**, the rotational speed is reduced, with the rotational speed of the first voltage

control member **71** being less than that of the first drive force receiving member **12**. This effectively controls the rotational speed of the first voltage control member **71** to enable effective control of circuit making and breaking by the first voltage control member **71** and improve detection precision.

(74) In this embodiment, the detectable device **70** further comprises a driving force cut-off mechanism for cutting off driving force transmitted to the first voltage control member **71**. Specifically, the driving force cut-off mechanism may be a first missing tooth portion disposed on the first main body portion **710** of the first voltage control member **71**. When the first missing tooth portion rotates to face the second gear portion **822** of the second transmission member **82**, driving force transmission is cut off, the first voltage control member **71** stops rotating, and detection is complete. In some embodiments, the first voltage control member **71** is configured to move along its rotation axis. When detection is complete, the first voltage control member **71** is configured to move along its rotation axis to disengage the second gear portion **822** of the second transmission member **82** from the first drive force receiving portion **711** of the first voltage control member **71**.

(75) This embodiment of the developing cartridge **10** also comprises a developer supply port **84** disposed at the second end portion **102** of the developing cartridge **10**. The developer supply port **84** does not overlap with the second end cover **104** disposed at the second end portion **102** of the developing cartridge **10**. The developer supply port **84** is disposed at the second end portion **102** of the developing cartridge **10** to facilitate developer filling. Since a plurality of transmission members are disposed at the first end portion **101** of the developing cartridge **10**, placing the developer supply port **84** at the first end portion **101** would not facilitate developer filling.

(76) When the developing cartridge **10** is installed in the imaging device P, the first conductive member **74**/first force applying portion **105** abuts against the switching member **503** of the imaging device P, moving the switching member **503** from its initial position to the closed position (as shown in FIG. **13**). The first conductive member **74** electrically connects with the switching member **503**, the fourth conductive member **77** electrically connects with the grounding terminal K, and the third conductive member **76** and fourth conductive member **77** are not in contact. At this time, the detection unit **501** detects a voltage value of U**1**.

(77) The detection process for the developing cartridge is described with reference to FIGS. **12-14**.

(78) When detection of the developing cartridge **10** begins, the first drive force receiving member **12** receives driving force and rotates, causing the first voltage control member **71** to receive driving force and rotate. As the first voltage control member **71** rotates, the first driving portion **712** drives the first force receiving portion **771** of the fourth conductive member **77** to make electrical contact with the third conductive member **76**, at which point the voltage value of the detection device **50** becomes U**2**. The values of U**2** and U**1** are different, and the detection unit **501** detects and records this voltage change.

(79) In this embodiment, the first voltage control member **71** comprises a plurality of first driving portions **712**, allowing the third conductive member **76** and fourth conductive member **77** to switch multiple times between electrically connected and non-connected states. The detection unit **501** detects multiple voltage value changes to identify information such as the capacity and specifications of the developing cartridge **10**. When detection is complete, driving force to the first voltage control member **71** is cut off, it no longer receives driving force, the first driving portion **712** no longer applies force to the fourth conductive member **77**, and the third conductive member **76** and fourth conductive member **77** remain in an electrically non-connected state. The absolute value of U**1** is greater than the absolute value of U**2**, with U**2** preferably being 0.

(80) In some embodiments, impedance elements may be included in the conductive assembly.

(81) In some embodiments, the first conductive member **74** and second conductive member **75** may be metal wires, metal conductive plates, conductive rubber, conductive resin, or similar materials.

(82) In some embodiments, driving force may be transmitted to the first voltage control member **71** by utilizing a torsion spring or spring that has accumulated elastic potential energy without needing

to receive driving force from the first drive force receiving member **12**.

(83) In some embodiments, driving force may be transmitted to the first voltage control member **71** through the second drive force receiving member **203** disposed in the drum cartridge.

(84) In some embodiments, the second conductive member may be an independently disposed conductive member.

(85) The developing cartridge **10** having the above structure provides a novel detection structure with improved detection precision and simplified construction.

(86) The developing cartridge **10** having the above structure eliminates the need for mechanisms to strike the switching member/swing arm, reducing damage to the swing arm. It also reduces vibration from striking actions, reducing noise and improving user experience.

Embodiment 2

(87) This embodiment represents an advancement based on Embodiment 1. Common elements between the embodiments will not be repeated; only the substantive differences will be described. The specification drawings from the previous embodiment may be referenced for identical components.

(88) As illustrated in FIG. **15**, the conductive assembly of the developing cartridge **10** in this embodiment is configured to establish electrical connection with the first inner wall P**21** of the door P**2** of the imaging device P. Specifically, the first electrical connection portion **73***a* extends from the rear end of the developing cartridge **10**. When the door P**2** is in its closed position, the first inner wall P**21** of the door P**2** establishes electrical connection with the first electrical connection portion **73***a*. In this particular embodiment, the first electrical connection portion **73***a* extends outward from the first end cover **103**.

(89) The first electrical connection portion **73***a* is preferably configured to be elastically deformable, enhancing the stability of the contact with the door P**2**.

(90) In certain embodiments, the first electrical connection portion **73***a* is configured to be movable relative to the developing frame **11**, thereby preventing any potential interference with the door P**2**.

(91) The imaging device P may, in some embodiments, be configured to accommodate multiple developing cartridges, with each cartridge incorporating at least one first electrical connection portion **73***a*. These multiple first electrical connection portions **73***a* are designed with varying protrusion lengths from their respective developing frames.

(92) This structural configuration of the developing cartridge **10** achieves enhanced contact stability with the door P**2**. Moreover, compared to configurations where the first electrical connection portion **73***a* contacts the first upper wall P**41**, this design eliminates potential contact reliability issues that might arise from contact with labels P**47**.

(93) The grounding terminal K may be at least one of the following elements: the ground terminal **142** of the storage unit **14**, the imaging device contact pin that contacts the ground terminal **142**, the inner wall of the door P**2**, the first upper wall P**41**, and the grounding conductive members P**311**.

Embodiment 3

(94) This embodiment represents a further advancement based on either Embodiment 1 or 2. Common elements shared among the embodiments will not be repeated; only the substantive differences will be detailed. The specification drawings from previous embodiments may be referenced for identical components.

(95) As shown in FIGS. **16** and **17**, the first electrical connection portion **73***b* is configured to move between a first position and a second position under external force application. In the first position, the first electrical connection portion **73***b* protrudes relative to the developing frame **11** and can establish electrical connection with the grounding terminal K. In the second position, the first electrical connection portion **73***b* retracts relative to the first position, preferably retracting relative to the developing frame **11**. The movement of the first electrical connection portion **73***b* may comprise sliding, pivoting, or rotating motion.

(96) In this embodiment, the first electrical connection portion **73***b* additionally comprises a second

force receiving portion **731** capable of receiving an actuating force from the imaging device P to initiate movement, specifically receiving an actuating force from the second surface P**44**.

(97) The first electrical connection portion **73***b* is preferably configured to move in the front-rear direction of the developing cartridge **10**. As illustrated in FIGS. **16** and **17**, when the developing cartridge **10** is not installed in the imaging device P, the first electrical connection portion **73***b* occupies the first position and does not protrude beyond the rear end of the developing cartridge **10**. As shown in FIG. **17**, upon installation of the developing cartridge **10** in the imaging device P, the first electrical connection portion **73***b* abuts against the second surface P**44** of the second upper wall P**42** of the imaging device P, causing a portion of the first electrical connection portion **73***b* to protrude beyond the rear end of the developing cartridge **10**. When the door P**2** is subsequently closed, the first electrical connection portion **73***b* can establish electrical connection with the first inner wall P**21** of the door P**2** to achieve grounding.

(98) This structural configuration of the developing cartridge **10** allows the first electrical connection portion **73***b* to retract within the developing frame **11** during packaging operations, preventing potential damage during packaging and transportation processes.

(99) In certain embodiments, the second force receiving portion **731** may be configured to be directly or indirectly driven by the first drive force receiving member **12**, enabling movement of the first electrical connection portion **73***b* between the first position and second position.

(100) The grounding terminal K may be at least one of the following elements: the ground terminal **142** of the storage unit **14**, the imaging device contact pin that contacts the ground terminal **142**, the inner wall of the door P**2**, the first upper wall P**41**, and the grounding conductive members P**311**.

Embodiment 4

(101) This embodiment represents a further advancement based on Embodiments 1, 2, or 3. Common elements will not be repeated; only the substantive differences will be detailed. The specification drawings from previous embodiments may be referenced for identical components.

(102) As shown in FIGS. **18-22**, in this embodiment, the first voltage control member **71** governs the selective contact between the conductive assembly **72***c* and the grounding terminal K. Specifically, the first electrical connection portion **73***c* of the conductive assembly **72***c* can transition between states of electrical contact and non-contact with the grounding terminal.

(103) The conductive assembly **72***c* in this embodiment comprises first conductive member **74***c*, second conductive member **75***c*, third conductive member **76***c*, and fourth conductive member **77***c*. The first conductive member **74***c* maintains electrical connection with the second conductive member **75***c* and can establish electrical connection with switching member **503** upon installation in the imaging device P. The second conductive member **75***c* extends from the first end portion **101** to the second end portion **102** of the developing cartridge **10**, maintaining electrical connection with the third conductive member **76***c*. The third conductive member **76***c* establishes electrical connection with the fourth conductive member **77***c*, which in turn can connect with the grounding terminal K. Preferably, the third conductive member **76***c* and fourth conductive member **77***c* are either integrally formed or fixedly connected.

(104) The first electrical connection portion **73***c* of the fourth conductive member **77***c* is configured for movement between the first position and second position. In the first position, the first electrical connection portion **73***c* protrudes relative to the developing frame **11** or first end cover **103**, enabling electrical connection with the grounding terminal. In the second position, the first electrical connection portion **73***c* retracts relative to the first position, preferably retracting relative to the developing frame **11**. The movement of the first electrical connection portion **73***c* may comprise sliding, pivoting, or rotating motion. In this embodiment, the first electrical connection portion **73***c* is slidably mounted on the first end portion **101**/first end cover **103**, with the first end cover **103** incorporating a receiving cavity **731***c* that accommodates at least a portion of the first electrical connection portion **73***c*. The first electrical connection portion **73***c* comprises a third force receiving portion **732***c*. During rotation of the first voltage control member **71**, the first driving

portion **712** can apply force to the third force receiving portion **732***c* of the first electrical connection portion **73***c*, causing movement of the first electrical connection portion **73***c* from the second position to the first position to establish electrical connection with the grounding terminal K. Preferably, the first electrical connection portion **73***c* is configured for movement in the front-rear direction of the developing cartridge **10**.

(105) The detectable device further comprises a first reset member **733***c* configured to urge the first electrical connection portion **73***c* to move from the first position to the second position. For example, when the first driving portion **712** ceases applying force to the force receiving portion of the first electrical connection portion **73***c*, the first electrical connection portion **73***c* moves from the first position to the second position under the force of the first reset member **733***c*. Preferably, the third conductive member **76***c* serves as the first reset member **733***c*, configured with a torsion spring structure. It accumulates elastic potential energy when the first electrical connection portion **73***c* moves from the second position to the first position. When the first driving portion **712** no longer applies force to the force receiving portion of the first electrical connection portion **73***c*, the third conductive member **76***c* releases this stored elastic potential energy, enabling the first electrical connection portion **73***c* to move from the first position to the second position.

(106) Referring to FIGS. **20**-**22**, when the developing cartridge **10** has not been installed in the imaging device P, as shown in FIG. **20**, the switching member **503** remains in its non-conducting open position. Upon installation of the developing cartridge **10** in the imaging device P, the swing arm **504** receives force from the first force applying portion **105**, causing the switching member **503** to transition from its open position to the closed position. At this point, while the detectable device is not conducting with the grounding terminal K, the detection unit **501** registers a voltage value of U**1**.

(107) When the developing cartridge **10** initiates its detection sequence, as illustrated in FIG. **22**, the first driving portion **712** actuates the first electrical connection portion **73***c* to move from the second position to the first position. The first electrical connection portion **73***c* establishes contact with the grounding terminal, at which point the detection unit **501** registers a voltage value of U**2**. When the first driving portion **712** ceases to drive the first electrical connection portion **73***c*, the first electrical connection portion **73***c* returns from the first position to the second position due to the action of the first reset member **733***c*, breaking contact of the first electrical connection portion **73***c* with the grounding terminal K. The detection unit **501** then registers a voltage value of U**1**. The values U**1** and U**2** are distinct, enabling the detection unit **501** to detect these voltage signal variations and thereby determine information about the developing cartridge **10**.

(108) The developing cartridge **10** with this structural configuration achieves more stable electrical connections of the detectable assembly.

(109) The grounding terminal K may be at least one of the following elements: the ground terminal **142** of the storage unit **14**, the imaging device contact pin that contacts the ground terminal **142**, the inner wall of the door P**2**, the first upper wall P**41**, and the grounding conductive members P**311**.

Embodiment 5

(110) This embodiment represents a further advancement based on Embodiments 1, 2, 3, or 4. Common elements will not be repeated; only the substantive differences will be detailed. The specification drawings from previous embodiments may be referenced for identical components.

(111) As illustrated in FIGS. **23**-**29**, in this embodiment, the first voltage control member **71***d* controls selective contact between the conductive assembly **72***d* and the switching member **503**. Specifically, the first conductive member **74***d* can transition between states of electrical contact and non-contact with the grounding terminal K.

(112) In this embodiment, the first conductive member **74***d* comprises an electrical contact portion **741***d* capable of establishing electrical contact with the switching member **503**. In this embodiment, the first conductive member **74***d* comprises separately formed third sub-member **742***d* and second sub-member **743***d*. The third sub-member **742***d* comprises a third sub-gear portion **744***d* and the

electrical contact portion **741***d*, while the second sub-member **743***d* comprises a second sub-gear portion **745***d*. The third sub-gear portion **744***d* meshes with the second sub-gear portion **745***d*, with both the third sub-member **742***d* and second sub-member **743***d* being fabricated from conductive materials.

(113) The second conductive member **75***d* establishes electrical connection with the first conductive member **74***d*, specifically connecting with the second sub-member **743***d*. Preferably, the second conductive member **75***d* is configured as an agitation member shaft.

(114) The third conductive member **76***d* maintains electrical connection with the second conductive member **75***d*. Specifically, in this embodiment, the third conductive member **76***d* is replaced by the electrically conductive second transmission member **82**. The third conductive member **76***d* incorporates a fifth gear portion **821** and a second gear portion **822**, with the fifth gear portion **821** receiving driving force from the first drive force receiving member **12**/first transmission member **81**, and the second gear portion **822** transmitting this driving force onward. The fourth conductive member **77***d* establishes electrical connection with the third conductive member **76***d*. Specifically, the fourth conductive member **77***d* comprises a first sub-member **771***d* and first electrical connection portion **73***d*. In this embodiment, the first sub-member **771***d* incorporates a first sub-gear portion **772***d* and a notched portion **773***d*, with the first sub-gear portion **772***d* meshing with the second gear portion **822** to receive driving force. The first electrical connection portion **73***d* is configured to move between the first position and second position. In the first position, the first electrical connection portion **73***d* can establish electrical connection with the grounding terminal K, while in the second position, the first electrical connection portion **73***d* cannot establish such connection with the grounding terminal K.

(115) In this embodiment, the first electrical connection portion **73***d* is configured for translational motion between the first and second positions. The first electrical connection portion **73***d* is configured as a rod member, with one end abutting against the end face of the first sub-member **771***d* and the other end configured for contact with the grounding terminal K. The first electrical connection portion **73***d* is slidably mounted on the first end cover **103** in this embodiment.

(116) The detectable device further comprises a first elastic member **79***d* that ensures the first electrical connection portion **73***d* abuts against the end face of the first sub-member **771***d*. One end of the first elastic member **79***d* connects to the first electrical connection portion **73***d*, while the other end abuts against the first end cover **103**. When the first electrical connection portion **73***d* enters the notched portion **773***d*, the force from the first elastic member **79***d* enables movement of the first electrical connection portion **73***d* from the first position to the second position. In the first position, the first electrical connection portion **73***d* can contact the grounding terminal K, while in the second position, it retracts and breaks contact with the grounding terminal K.

(117) The first voltage control member **71***d* is disposed at the second end portion **102** of the developing cartridge **10**. It comprises a non-electrical contact portion **711***d* positioned between adjacent electrical contact portions **741***d* of the first conductive member **74***d*, creating an alternating arrangement of electrical contact portions **741***d* and non-electrical contact portions **711***d*. When the switching member **503** makes contact with an electrical contact portion **741***d*, the conductive assembly **72***d* establishes electrical connection with the switching member **503**. When the switching member **503** makes contact with a non-electrical contact portion **711***d*, the conductive assembly **72***d* breaks electrical connection with the switching member **503**. The design may incorporate multiple electrical contact portions **741***d* and non-electrical contact portions **711***d* arranged in an alternating sequence.

(118) Referring to FIGS. **27-29**, when the developing cartridge **10** is not installed in the imaging device P, as shown in FIG. **27**, the switching member **503** remains in its non-conducting open position. Upon installation of the developing cartridge **10** in the imaging device P, the swing arm receives force from the first force applying portion **105**, causing the switching member **503** to move from its open position to the closed position. At this point, the first electrical connection

portion **73**_d_ is in the first position maintaining electrical contact with the grounding terminal K, while the switching member **503** contacts the non-electrical contact portion **711**_d_ of the first voltage control member **71**_d_ (with the first voltage control member **71**_d_ in its third position). The conductive assembly **72**_d_ remains in a state of non-electrical connection with the switching member **503**, and the detection unit **501** registers a voltage value of U**1**.
(119) When detection of the developing cartridge **10** begins, as shown in FIG. **29**, the first drive force receiving member **12** receives and transmits driving force to the first conductive member **74**_d_. As the first conductive member **74**_d_ and first voltage control member **71**_d_ rotate, the electrical contact portion **741**_d_ of the first conductive member **74**_d_ contacts the switching member **503** (with the first voltage control member **71**_d_ now in the fourth position). The conductive assembly **72**_d_ establishes conduction with the switching member **503**, the detection device is grounded, and the detection unit **501** registers a voltage value of U**2**. As rotation of the first conductive member **74**_d_ and first voltage control member **71**_d_ continues and the non-electrical contact portion **711**_d_ again contacts the switching member **503**, the conductive assembly **72**_d_ breaks electrical connection with the detection device, and the detection unit **501** returns to registering a voltage value of U**1**. Similarly, the electrical contact portion **741**_d_ and non-electrical contact portion **711**_d_ alternately make electrical contact with the switching member **503**, enabling the detection unit **501** to detect the alternating changes in voltage and thereby determine information about the developing cartridge **10**.
(120) The developing cartridge **10** with this structural configuration achieves more stable electrical connections of the detectable assembly.
(121) The grounding terminal K may be at least one of the following elements: the ground terminal **142** of the storage unit **14**, the imaging device contact pin that contacts the ground terminal **142**, the inner wall of the door P**2**, the first upper wall P**41**, and the grounding conductive members P**311**.
Embodiment 6
(122) This embodiment represents an advancement based on Embodiments 1-5. Common elements between the embodiments will not be repeated; only the substantive differences will be described. The specification drawings from previous embodiments may be referenced for identical components.
(123) As illustrated in FIGS. **30-32**, in this embodiment, the conductive assembly achieves grounding through electrical connection with the ground terminal **142** of the storage unit **14**. Specifically, the fourth conductive member **77**_e_ establishes electrical connection with the ground terminal **142** of the storage unit **14**, achieving grounding through the connection between the ground terminal **142**/fourth conductive member **77**_e_ and the contact pin of the imaging device P.
(124) As shown in the figures, this embodiment further comprises a second voltage control member **9**, which controls the voltage/current value of the developing roller **13**. Preferably, the second voltage control member **9** controls the circuit voltage/current value through making and breaking the circuit. The second voltage control member **9** can receive external force to move between a fifth position and a sixth position. In the fifth position, the developing roller establishes electrical connection with the grounding terminal, while in the sixth position, the developing roller is not electrically connected to the grounding terminal. In this embodiment, the second conductive member **75**_e_ is the developing roller **13**.
(125) The first conductive member **74**_e_ maintains electrical connection with the developing roller **13**, which in turn electrically connects to the third conductive member **76**_e_, and the third conductive member **76**_e_ electrically connects to the fourth conductive member **77**_e_. The first conductive member **74**_e_ can establish electrical connection with the power supply electrode of the imaging device, which supplies electrical power to the developing roller. Specifically, this power supply electrode may be the switching member **503**. In some embodiments, the power supply electrode may be configured as an independently disposed component.
(126) Preferably, the second voltage control member **9** is configured as a movable member capable

of movement between the fifth and sixth positions. In the fifth position, the second voltage control member **9** drives the third conductive member **76***e* and/or fourth conductive member **77***e* to establish an electrically connected state between the third conductive member **76***e* and the fourth conductive member **77***e*. In the sixth position, the second voltage control member **9** does not drive the third conductive member **76***e* and/or fourth conductive member **77***e*, leaving the third conductive member **76***e* and the fourth conductive member **77***e* in an electrically non-connected state.
(127) The second voltage control member **9** is configured to move in response to movement of the separation force applying member of the imaging device P or the separation force transmission member **206** of the drum cartridge. It can directly or indirectly drive at least a portion of the third conductive member **76***e* and/or fourth conductive member **77***e*, enabling switching between contact and non-contact states between the third conductive member **76***e* and the fourth conductive member **77***e*. In other words, the third conductive member **76***e* can switch between contact and non-contact states with the fourth conductive member **77***e*.
(128) The second voltage control member **9** can be rotatably connected to the developing frame **11** either directly or indirectly. In this embodiment, the second voltage control member **9** is rotatably mounted at the first end portion **101** of the developing cartridge **10**. The second voltage control member **9** is configured as a rotary member comprising a third main body portion **90**, second force receiving portion **92**, and second driving portion **91**. The second force receiving portion **92** is disposed on the third main body portion **90** to receive actuating force from the separation force applying member or separation force transmission member **206**, enabling rotation of the second voltage control member **9**. The second driving portion **91**, also mounted on the third main body portion **90**, can drive at least a portion of the third conductive member **76***e* and/or fourth conductive member **77***e* to establish electrical contact between the third conductive member **76***e* and the fourth conductive member **77***e*.
(129) During imaging operations of the developing cartridge **10**, the first conductive member **74***e* establishes electrical connection with the switching member **503** and receives electrical power. At this time, the developing roller **13** receives a voltage value of U**3**, while the ground terminal **142** of the storage unit **14** maintains electrical connection with the contact pin of the imaging device P. The third conductive member **76***e* and fourth conductive member **77***e* remain in a non-contact state.
(130) When the developing cartridge **10** executes a photosensitive drum **201** cleaning operation, the separation force applying member or separation force transmission member **206** applies force to the second voltage control member **9**, causing it to rotate. As the second voltage control member **9** rotates, the second driving portion **91** drives the third conductive member **76***e* into electrical contact with the fourth conductive member **77***e*, at which point the developing roller **13** voltage value changes to U**4**. The values U**3** and U**4** differ; when the voltage is U**3**, developer from the developing roller **13** can transfer to the photosensitive drum **201** surface; however, at voltage U**4**, developer from the developing roller **13** does not transfer to the photosensitive drum **201**, thereby facilitating cleaning of the photosensitive drum **201** without developer accumulation during the cleaning process. The absolute value of U**3** exceeds that of U**4**, with U**4** preferably being 0.
(131) Upon completion of cleaning, when the separation force applying member or separation force transmission member **206** no longer applies force to the second voltage control member **9**, the second voltage control member **9** rotates in the reverse direction under the force of a second elastic reset member, returning the third conductive member **76***e* and fourth conductive member **77***e* to an electrically non-connected state.
(132) This structural configuration of the developing cartridge **10** reduces the risk of developer adhesion during photosensitive drum **201** cleaning operations, thereby improving cleaning effectiveness. Furthermore, this cartridge structure eliminates the need for complex separation mechanisms and does not require separation between the photosensitive drum and developing roller.

(133) In certain embodiments, the conductive assembly **72***e* may be configured to achieve grounding through electrical connection with at least one of: the first upper wall P**41**, second upper wall P**42**, or door P**2**'s inner wall of the imaging device P.
(134) In some embodiments, the first voltage control member **71***d* and second voltage control member **9** may be configured as components that adjust circuit resistance, controlling circuit voltage/current values through resistance modification.
(135) Throughout these embodiments and their variations, “driving” encompasses both direct and indirect driving mechanisms, while “connection” encompasses both direct and indirect connections.
(136) Some embodiments may feature a developing cartridge without a first voltage control member, incorporating only the second voltage control member **9**.
(137) Conversely, other embodiments may include the first voltage control member while omitting the second voltage control member **9**.
(138) In certain embodiments, the second voltage control member **9** may modify the developing roller's voltage by controlling the electrical contact and non-contact between the conductive assembly and at least one of the first upper wall P**41**, second upper wall P**42**, or door P**2**'s inner wall of the imaging device P. Specifically, this voltage modification of the developing roller may be achieved through the contact and non-contact between the fourth conductive member **77***e* and the aforementioned grounding terminals.
(139) This embodiment may be integrated with Embodiments 1-5, and the second voltage control member may be incorporated into any of Embodiments 1-5.
(140) The electrical connections described in the present disclosure may be implemented as either direct or indirect electrical connections.
Embodiment 7
(141) This embodiment represents a further advancement of Embodiment 4. Common elements between the embodiments will not be repeated; only substantive differences will be detailed. The specification drawings from previous embodiments may be referenced for identical components.
(142) As shown in FIGS. **1-5***b* and FIGS. **33-36**, in this embodiment, the conductive assembly can establish electrical connection with the switching member, while being capable of transitioning between electrically connected and non-connected states with the grounding terminal. Specifically, the first voltage control member **71***f* governs selective contact between the conductive assembly and grounding terminal K, enabling the first electrical connection portion **73***f* of the conductive assembly to switch between states of electrical contact and non-contact with grounding terminal K.
(143) The conductive assembly in this embodiment comprises first conductive member **74***f*, second conductive member **75***f*, third conductive member **76***f*, and fourth conductive member **77***f*. The first conductive member **74***f* maintains electrical connection with the second conductive member **75***f* and can establish electrical connection with switching member **503** upon installation in imaging device P. In this embodiment, the first conductive member **74***f* is fabricated from conductive metal.
(144) The second conductive member **75***f* extends from the first end portion **101** to the second end portion **102** of the developing cartridge **10**, maintaining electrical connection with the third conductive member **76***f*. It may incorporate at least one of the following components: developing roller shaft **132**, developing roller **13**, supply roller **15**, doctor blade **25**, or may at least include an independent conductive component separate from developing roller **13**, supply roller **15** and doctor blade **25**. In this embodiment, the doctor blade **25** serves as the second conductive member **75***f*.
(145) The third conductive member **76***f* is positioned at the first end portion **101** of the developing cartridge **10** and maintains electrical connection with the fourth conductive member **77***f*, which in turn can connect to grounding terminal K. Preferably, the third conductive member **76***f* and fourth conductive member **77***f* are either integrally formed or fixedly connected.
(146) In certain embodiments, the third conductive member **76***f* is configured to remain within the interior of the first end cover **103***f* or developing frame **11**, preventing protrusion beyond the exterior of the developing cartridge from the first end cover **103***f* or developing frame **11**. This

configuration design protects the third conductive member **76***f* from potential damage caused by external component interference. More specifically, the third conductive member **76***f* incorporates multiple bend portions to navigate around blocking structures present on the first end cover **103***f* or developing frame **11**.

(147) The first electrical connection portion **73***f* of the fourth conductive member **77***f* is configured to move between a first position and a second position. In the first position, the first electrical connection portion **73***f* protrudes relative to the developing frame **11** or first end cover **103***f*, enabling electrical contact with grounding terminal K in the first position. In the second position, the first electrical connection portion **73***f* retracts relative to the first position, preferably retracting relative to the developing frame **11** or first end cover **103***f*. The first electrical connection portion **73***f* features elastic deformability, enabling resilient electrical contact with grounding terminal K (door), thereby enhancing contact stability.

(148) In this embodiment, the third conductive member **76***f* can function as the first reset member **733***f*, forcing the first electrical connection portion **73***f* to move from the first position to the second position, that is, the third conductive member **76***f* can be considered as the first reset member **733***f*. The third conductive member **76***f* incorporates first connection end **761**, second connection end **762**, and elastic portion **763**. The first connection end **761** connects electrically to the second conductive member **75***f*, while the second connection end **762** connects electrically to the fourth conductive member **77***f*. The elastic portion **763** connects the first connection end **761** to the second connection end **762**. The elastic portion **763** stretches when the first electrical connection portion **73***f* moves from the second position to the first position, and its elastic recovery force returns the first electrical connection portion **73***f* from the first position to the second position. Specifically, when the first electrical connection portion **73***f* moves from the second position to the first position under the drive of the first voltage control member **71***f*, the elastic portion **763** stretches. When the first voltage control member **71***f* ceases driving the first electrical connection portion **73***f* (whether directly or indirectly), the elastic portion **763** returns to its original state, pulling the first electrical connection portion **73***f* back to the second position from the first position.

(149) Viewed from the front-rear direction of the developing cartridge **10**, the first electrical connection portion **73***f* at least partially overlaps with the first voltage control member **71***f*. Along the left-right direction of the developing cartridge **10**, the first electrical connection portion **73***f* is positioned between the sixth gear portion **124** and the electrical contacts of storage unit **148**. Preferably, the first electrical connection portion **73***f* is positioned between the sixth gear portion **124** and the first force receiving portion **121**.

(150) The first end cover **103***f* of this embodiment covers at least a portion of the fourth conductive member **77***f* and features an opening portion **1031** and an accommodation cavity **1032**. The opening portion **1031**, situated on the rear side of the first end cover **103***f* and facing toward the rear of the developing cartridge **10**, allows the first electrical connection portion **73***f* to extend outward. When the first electrical connection portion **73***f* occupies the first position, at least part of it extends beyond the opening portion **1031**. In the second position, the first electrical connection portion **73***f* remains within the opening portion **1031**. The accommodation cavity **1032**, located between storage unit **148** and developing frame **11**, accommodates the fourth conductive member **77***f*.

(151) This structural configuration, where the first end cover **103***f* covers at least a portion of the fourth conductive member **77***f*, provides enhanced protection for the fourth conductive member **77***f* by the first end cover **103***f*. Additionally, during packaging operations of the developing cartridge **10**, the first electrical connection portion **73***f* of the first conductive member **74***f* is in the second position, which can avoid the first electrical connection portion **73***f* puncturing the packaging material, thereby reducing the difficulty and cost of packaging.

(152) The developing cartridge **10** further incorporates a movable member **9***f* that supports at least a portion of either the fourth conductive member **77***f* or the first electrical connection portion **73***f*.

This movable member **9***f* can move relative to the developing frame **11** of the developing cartridge **10**. In this embodiment, the movable member **9***f* is specifically configured to slide in the front-rear direction relative to the developing frame **11**, preferably mounted in a sliding configuration on the first end cover **103***f*. Alternative embodiments may position the movable member **9***f* on the developing frame **11**. The movable member **9***f* may also move in either left-right or up-down directions, provided it can effectively switch between contact and non-contact states with grounding terminal K.
(153) The movable member **9***f* comprises a connection portion **91***f*, a first accommodation portion **92***f*, a force receiving portion **93***f* and a second force applying portion **94***f*. The connection portion **91***f* enables sliding connection with the connected portion **1033** of the first end cover **103***f*. Specifically, at least one of the connection portion **91***f* and the connected portion **1033** functions as a sliding block, while the other one serves as a corresponding sliding groove.
(154) The first accommodation portion **92***f* is configured to accommodate at least a portion of the fourth conductive member **77***f*, with the first electrical connection portion **73***f* able to extend through the opening of the first accommodation portion **92***f*. The force receiving portion **93***f* receives actuating force from the first voltage control member **71***f* to enable sliding movement of the movable member **9***f*. This force receiving portion **93***f* extends toward the developing roller **13**, terminating in a force receiving surface **931***f*. The force receiving surface **931***f* is configured as an inclined plane facing both downward and forward. The second force applying portion **94***f* applies force to the fourth conductive member **77***f* to facilitate movement of the first electrical connection portion **73***f* from the second position to the first position. Specifically, the fourth conductive member **77***f* abuts against the second force applying portion **94***f*.
(155) In this embodiment, the first driving portion **712** of the first voltage control member **71***f* incorporates an arcuate drive surface **7121**. This arcuate drive surface **7121** works in conjunction with the force receiving surface **931***f* to enable sliding movement of the movable member **9***f*. The cooperation between the arcuate drive surface **7121** and the force receiving surface **931***f* ensures smoother driving action.
(156) In this embodiment, the developing cartridge **10** also features a reset mechanism that allows users to return the detectable device to its initial state before detection. In this embodiment, the reset mechanism comprises a first indicator position **871***f* provided on the first voltage control member **71***f* and a second indicator position **872***f* provided on the first end cover **103***f*. When the first indicator position **871***f* and the second indicator position **872***f* are aligned, the first voltage control member **71***f* is in its initial position, and the detectable device is in its initial state before detection. Specifically, the first indicator position **871***f* is a first surface provided on the first driving portion **712**, while the second indicator position **872***f* is a second surface provided on the upper side of the end cover. When the first surface and the second surface are flush, the first voltage control member **71***f* is in its initial position, and the detectable device is in its initial state before detection.
(157) Referring to FIG. **20**, when the developing cartridge **10** is not installed in the imaging device P, the switching member **503** remains in its non-conducting open position. Referring to FIG. **21**, upon installation of the developing cartridge **10** in the imaging device P, the swing arm **504**, responding to force from the second force applying portion **94***f*, moves the switching member **503** from its open position to the closed position. At this stage, while the detectable device is not conducting with grounding terminal K, the detection unit registers a voltage value of U**1**.
(158) Referring to FIG. **22**, when detection of the developing cartridge **10** commences, the first driving portion **712** pushes the movable member **9***f* rearward. The second force applying portion **94***f* of the movable member **9***f* then applies force to the fourth conductive member **77***f*, causing the first electrical connection portion **73***f* to move rearward from the second position to the first position. As the first electrical connection portion **73***f* makes contact with grounding terminal K, the elastic portion **763** undergoes stretching, and the detection unit registers a voltage value of U**2**.
(159) When the first driving portion **712** ceases pushing the movable member **9***f*, the elastic

recovery force of elastic portion **763** acts on the fourth conductive member **77***f*, returning the first electrical connection portion **73***f* from the first position to the second position. Simultaneously, the movable member **9***f* moves forward under the force of the fourth conductive member **77***f*, breaking electrical contact between the first electrical connection portion **73***f* and grounding terminal K. The detection unit then registers a voltage value of U**1**. The values of U**1** and U**2** are different, and the detection unit detects the variation in the voltage signal. When the first driving portion **712** drives the movable member **9***f* again, the first electrical connection portion **73***f* and the grounding terminal K switch between contact and non-contact states once more. The detection unit detects the variation in voltage value to determine the information of the developing cartridge **10**. When users need to reset the detectable device, they only need to rotate the first voltage control member **71***f* to align the first surface with the second surface.
(160) The grounding terminal K may be at least one of: the storage unit's ground terminal, the imaging device's contact pin that connects with the ground terminal, the door P**2**'s inner wall, the device's side wall, the first upper wall P**41**, and the grounding conductive members P**311**.
(161) In certain embodiments, at least two of the first conductive member **74***f*, second conductive member **75***f*, third conductive member **76***f*, and fourth conductive member **77***f* may be integrally formed. When the first conductive member **74***f*, second conductive member **75***f*, third conductive member **76***f*, and fourth conductive member **77***f* are integrally formed, the conductive assembly may take the form of a single conductive wire, conductive metal wire, conductive resin or the like.
(162) In some embodiments, at least one of the first conductive member **74***f*, second conductive member **75***f*, third conductive member **76***f*, and fourth conductive member **77***f* may incorporate multiple separately configured conductive components.
(163) In certain embodiments, the conductive assembly supplies received electrical power to at least one of the developing roller, doctor blade, or supply roller; that is, the conductive assembly is electrically connected to at least one of the developing roller, doctor blade, and supply roller.
(164) In some embodiments, the conductive assembly is not electrically connected to at least one of the developing roller, the doctor blade, and the supply roller; the developing roller is electrically connected to the switching member **503** or the power supply component of the imaging device through an electrical connection component that is independent of the conductive assembly. The electrical connection component can make electrical contact with the switching member **503** without contacting the conductive assembly.
(165) In some embodiments, upon detection of the developing cartridge is complete, the fourth conductive member **77***f* abuts against the first voltage control member **71***f* to prevent the first voltage control member **71***f* from further rotating or reversing and engaging with the transmission member (such as the second transmission member **82**) again.
Embodiment 8
(166) This embodiment represents a further advancement of the technical solutions presented in Embodiments 1-7. Common elements will not be repeated, and relevant illustrations from previous embodiments may be referenced for identical components.
(167) In the process of implementing the technical solution, it was discovered through experimentation that, due to the elastic contact between the first electrical connection portion and the grounding terminal K, when the first electrical connection portion moves from a position where it is not in contact with the grounding terminal K to a position where it is in contact with the grounding terminal K (for example, when the first electrical connection portion moves from the second position to the first position), the first electrical connection portion may experience sliding or bouncing movements relative to the grounding terminal K. This can lead to unnecessary fluctuations in detection electrical signals, or poor contact quality, which could lead to detection failure.
(168) As illustrated in FIGS. **37** and **38**, this embodiment addresses the aforementioned technical challenge by incorporating an anti-slip portion **731***g* on the first electrical connection portion **73***g*.

This anti-slip portion **731***g* prevents sliding or bouncing movements of the first electrical connection portion **73***g* relative to grounding terminal K during contacting the grounding terminal K, thereby enhancing the stability of electrical contact between the first electrical connection portion **73***g* and grounding terminal K. This avoids unnecessary fluctuations in detection electrical signals or poor electrical contact, which could lead to detection failure.

(169) The first electrical connection portion **73***g* includes an anti-slip portion **731***g* and an mounting portion **732***g*. The mounting portion **732***g* is used for mounting the anti-slip portion **731***g*, and one end of the anti-slip portion **731***g* forms an assembly end **733***g*. The anti-slip portion **731***g* can be assembled onto the mounting portion **732***g* through the assembly end **733***g*. In this embodiment, the assembly end **733***g* is formed with a snap-fit structure **734***g*, which engages with the mounting portion **732***g* to prevent detachment of the anti-slip portion **731***g*.

(170) The mounting portion **732***g* is formed at one end of the fourth conductive member **77**. The anti-slip portion **731***g* is at least partially made of elastic conductive material, such as elastic conductive rubber. The elastic conductive rubber does not slip or bounce when it abuts against the grounding terminal K (e.g., the inner wall of the door, the side wall of the device, the first upper wall, and the grounding conductive member, etc.). The fourth conductive member **77** is configured as a spring, and when viewed along the axial direction of the fourth conductive member **77**, the anti-slip portion **731***g* covers the mounting portion **732***g*. In this embodiment, by providing the anti-slip portion **731***g*, the contact area between the first electrical connection portion **73***g* and the grounding terminal K can also be increased, thereby enhancing the stability of the electrical contact. Moreover, by using elastic conductive rubber as the anti-slip portion **731***g*, it can avoid puncturing the packaging bag when packaging the developing cartridge.

(171) Some embodiments feature anti-slip stripes on the end surface of the anti-slip portion **731***g* to increase frictional resistance.

(172) In some embodiments, the anti-slip portion **731***g* is made of other materials with anti-slip properties, such as conductive silicone, provided they enhance frictional force during contact with grounding terminal K.

(173) FIG. **39** illustrates a structural variation of the anti-slip portion **731***g* in a variant of this embodiment, where the assembly end **733***g* of the anti-slip portion **731***g* forms a chamber structure. This chamber structure enables the assembly end **733***g* to sleeve over the exterior of the mounting portion **732***g*, preventing detachment of the anti-slip portion **731***g*.

(174) The embodiments described above may be combined in various ways while remaining within the protective scope of the present disclosure.

(175) Finally, it should be noted that the above embodiments are intended to illustrate the technical solutions of the present disclosure rather than impose limitations. In order to distinguish different components, the present disclosure introduces terms such as “first” and “second”. These terms are not to be understood as quantity limitations. For example, the driving member described in the present disclosure has a sixth gear portion, but this does not necessarily mean that the driving member must have a first gear portion and a third gear portion, etc. According to the description in the specification, the components described as “first”, “second”, etc., may be singular or plural. Directional terms such as “upper”, “upper side”, “lower”, and “lower side” are based on the description in the accompanying drawings and are not specific limitations on their orientation. Although the present disclosure has been described in detail with reference to the foregoing embodiments, those of ordinary skill in the art should understand that they can still modify the technical solutions described in the foregoing embodiments, or make equivalent replacements of some or all of the technical features; and these modifications or replacements do not cause the essence of the corresponding technical solutions to depart from the scope of the technical solutions of the embodiments of the present disclosure.

## Claims

1. A developing cartridge, comprising: a developing frame internally provided with a developer chamber configured to accommodate developer; a developing roller rotatably disposed on the developing frame; a first drive force receiving member configured to receive a driving force for driving rotation of the developing roller; a first end cover, connected to the developing frame and at least partially covering the first drive force receiving member; a conductive assembly capable of switching internally between an electrically connected state and an electrically non-connected state, wherein the conductive assembly is provided with a first electrical connection portion that is elastically deformable; and a first voltage control member mounted on the first end cover and configured to move in response to movement of the first drive force receiving member, thereby switching the conductive assembly between the electrically connected state and the electrically non-connected state.
2. The developing cartridge according to claim 1, wherein the developing cartridge is detachably installable in an imaging device, the imaging device further comprises an openable door, the door has a groundable first inner wall, and the first electrical connection portion is configured to electrically contact the first inner wall.
3. The developing cartridge according to claim 2, wherein the first electrical connection portion comprises an anti-slip portion, the anti-slip portion abuts the first inner wall, and at least a portion of the anti-slip portion is made of an elastic conductive material.
4. The developing cartridge according to claim 1, wherein the first electrical connection portion is configured to move in a front-rear direction of the developing cartridge.
5. The developing cartridge according to claim 1, wherein the first electrical connection portion is configured to move between a first position and a second position under an external force; in the first position, the first electrical connection portion protrudes relative to the developing frame, and in the second position, the first electrical connection portion is retracted relative to the first position.
6. The developing cartridge according to claim 1, wherein the first electrical connection portion is disposed on the first end cover, and viewed from a front-rear direction of the developing cartridge, the first electrical connection portion at least partially overlaps with the first voltage control member.
7. The developing cartridge according to claim 1, wherein, along a longitudinal direction of the developing cartridge, the developing cartridge has a first end portion and an opposing second end portion, the conductive assembly comprises a first conductive member, a second conductive member, a third conductive member, and a fourth conductive member, the first conductive member is disposed at the second end portion, the second conductive member is electrically connected to the first conductive member, the third conductive member is electrically connected to the second conductive member and configured to connect to the fourth conductive member; at least a portion of the second conductive member extends between the first end portion and the second end portion of the developing cartridge, at least a portion of the third conductive member is disposed at the first end portion of the developing cartridge; the fourth conductive member includes the first electrical connection portion; the first voltage control member comprises a first main body portion, a first drive force receiving portion, and a first driving portion, the first drive force receiving portion is disposed on the first main body portion and configured to receive a driving force; the first driving portion is disposed on the first main body portion and is capable of driving the third conductive member or the fourth conductive member so that the third conductive member and the fourth conductive member is capable of switching between the electrically connected state and the electrically non-connected state.
8. The developing cartridge according to claim 7, wherein the first conductive member, the second

conductive member, the third conductive member and the fourth conductive member are separately formed or at least two are integrally formed.
9. The developing cartridge according to claim 7, wherein the fourth conductive member comprises a first sub-member and the first electrical connection portion, and the first sub-member is provided with a first sub-gear.
10. The developing cartridge according to claim 7, wherein the developing cartridge further comprises a supply roller rotatably supported on the developing frame and configured to supply developer to the developing roller, wherein the second conductive member is the supply roller.
11. A developing cartridge, comprising: a developing frame internally provided with a developer chamber configured to accommodate developer; a developing roller rotatably disposed on the developing frame; a first drive force receiving member configured to receive a driving force for driving rotation of the developing roller; a first end cover, connected to the developing frame and at least partially covering the first drive force receiving member; a conductive assembly capable of switching internally between an electrically connected state and an electrically non-connected state, wherein the conductive assembly is provided with a first electrical connection portion that is elastically deformable; and a first voltage control member configured to move in response to movement of the first drive force receiving member, thereby switching the conductive assembly between the electrically connected state and the electrically non-connected state.
12. The developing cartridge according to claim 11, wherein the developing cartridge is detachably installable in a drum cartridge having a photosensitive drum, the drum cartridge is detachably installable in an imaging device, the drum cartridge comprises a separation force transmission member, and the developing cartridge comprises a separation force receiving member configured to receive an acting force from the separation force transmission member or a separation force applying member of the imaging device to separate the developing roller from the photosensitive drum, before separation of the developing roller from the photosensitive drum, the first electrical connection portion has a first deformation amount, and after separation of the developing roller from the photosensitive drum, the first electrical connection portion has a second deformation amount, the first deformation amount is greater than the second deformation amount.
13. The developing cartridge according to claim 11, wherein the developing cartridge is detachably installable in a drum cartridge having a photosensitive drum, the drum cartridge is detachably installable in a main assembly of an imaging device, and the main assembly comprises a main frame and a mounting cavity configured to mount the developing cartridge, the main frame comprises a first upper wall and a second upper wall, the second upper wall is closer to an inside of the mounting cavity than the first upper wall, the first upper wall is made of a conductive material, the second upper wall is made of a non-conductive material, and the first electrical connection portion is configured to electrically connect with the first upper wall.
14. The developing cartridge according to claim 13, wherein the first upper wall comprises a plurality of labels, there is a first gap between the labels and the second upper wall, there is a second gap between the plurality of labels, and the first electrical connection portion makes electrical contact with the first gap or the second gap.
15. The developing cartridge according to claim 11, wherein the developing cartridge is detachably installable in a drum cartridge having a photosensitive drum, the drum cartridge is detachably installable in an imaging device, and the developing cartridge further comprises a storage unit having a ground terminal capable of electrically connecting with a contact pin of the imaging device, and the first electrical connection portion is capable of electrically connecting with the ground terminal.
16. The developing cartridge according to claim 11, wherein, along a longitudinal direction of the developing cartridge, the developing cartridge has a first end portion and an opposing second end portion, the conductive assembly comprises a first conductive member, a second conductive member, a third conductive member, and a fourth conductive member, the first conductive member

is disposed at the second end portion, the second conductive member is electrically connected to the first conductive member, the third conductive member is electrically connected to the second conductive member and configured to connect to the fourth conductive member; at least a portion of the second conductive member extends between the first end portion and the second end portion of the developing cartridge, at least a portion of the third conductive member is disposed at the first end portion of the developing cartridge; the fourth conductive member includes the first electrical connection portion; the first voltage control member comprises a first main body portion, a first drive force receiving portion, and a first driving portion, the first drive force receiving portion is disposed on the first main body portion and configured to receive a driving force; the first driving portion is disposed on the first main body portion and is capable of driving the third conductive member or the fourth conductive member so that the third conductive member and the fourth conductive member is capable of switching between the electrically connected state and the electrically non-connected state.

17. The developing cartridge according to claim 16, wherein the first conductive member, the second conductive member, the third conductive member and the fourth conductive member are separately formed or at least two are integrally formed.

18. The developing cartridge according to claim 16, wherein the imaging device comprises a detection device and a grounding terminal, the detection device comprises a detection unit, a power source and a switching member capable of forming a conductive circuit, the switching member is movable between a closed position and an open position, and the conductive assembly is capable of electrically connecting the switching member and the grounding terminal.

19. The developing cartridge according to claim 18, further comprising a first force applying portion disposed on the first conductive member or formed separately from the first conductive member, wherein the first force applying portion is configured to apply a force to the switching member when the developing cartridge is installed in the imaging device to move the switching member from an initial position to the closed position.

20. The developing cartridge according to claim 11, further comprising a deceleration mechanism disposed at a first end portion of the developing cartridge, wherein the deceleration mechanism comprises at least a first transmission member and a second transmission member, the first transmission member comprises a third gear portion and a fourth gear portion, a diameter of the third gear portion is smaller than that of the fourth gear portion, the second transmission member comprises a fifth gear portion and a second gear portion, the fourth gear portion meshes with a first gear portion of the first drive force receiving member, the third gear portion meshes with the fifth gear portion, and the second gear portion meshes with a first drive force receiving portion of the first voltage control member.